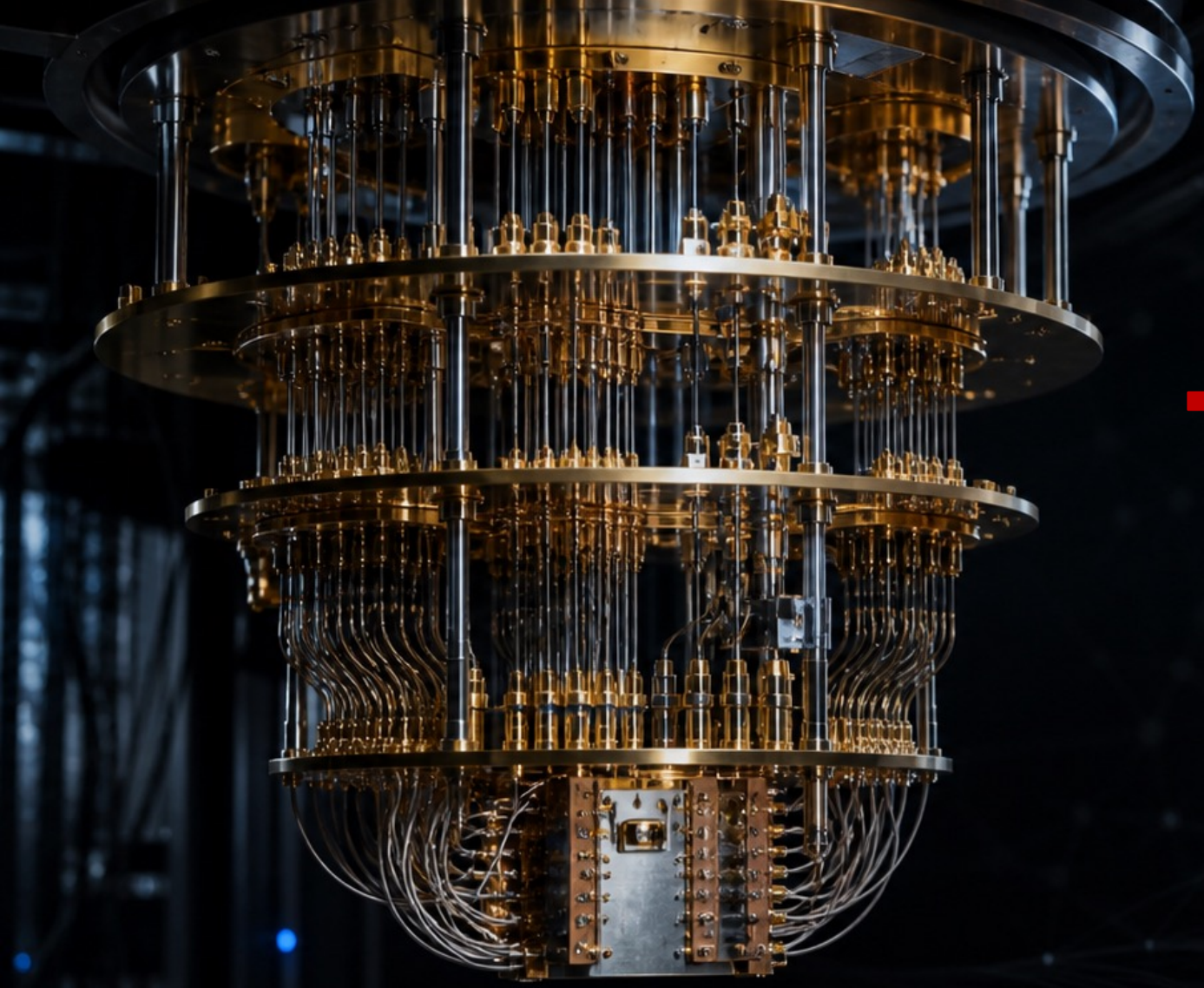


POST-QUANTUM CRYPTOGRAPHY

Protect Today
what must still be
Secret Tomorrow

{elysiumsecurity}
cyber protection & response

Version: 1.3

Author: Sylvain Martinez

Classification: PUBLIC

Presentation Overview

This presentation focuses on the impact future advances in computing will have on the cryptographic controls we have in place today.

Specifically, what do we need to do now to ensure we are safe tomorrow.

1. CONTEXT:

   WHY THE DEADLINE IS ALREADY UPON US

2. EXPOSURE:

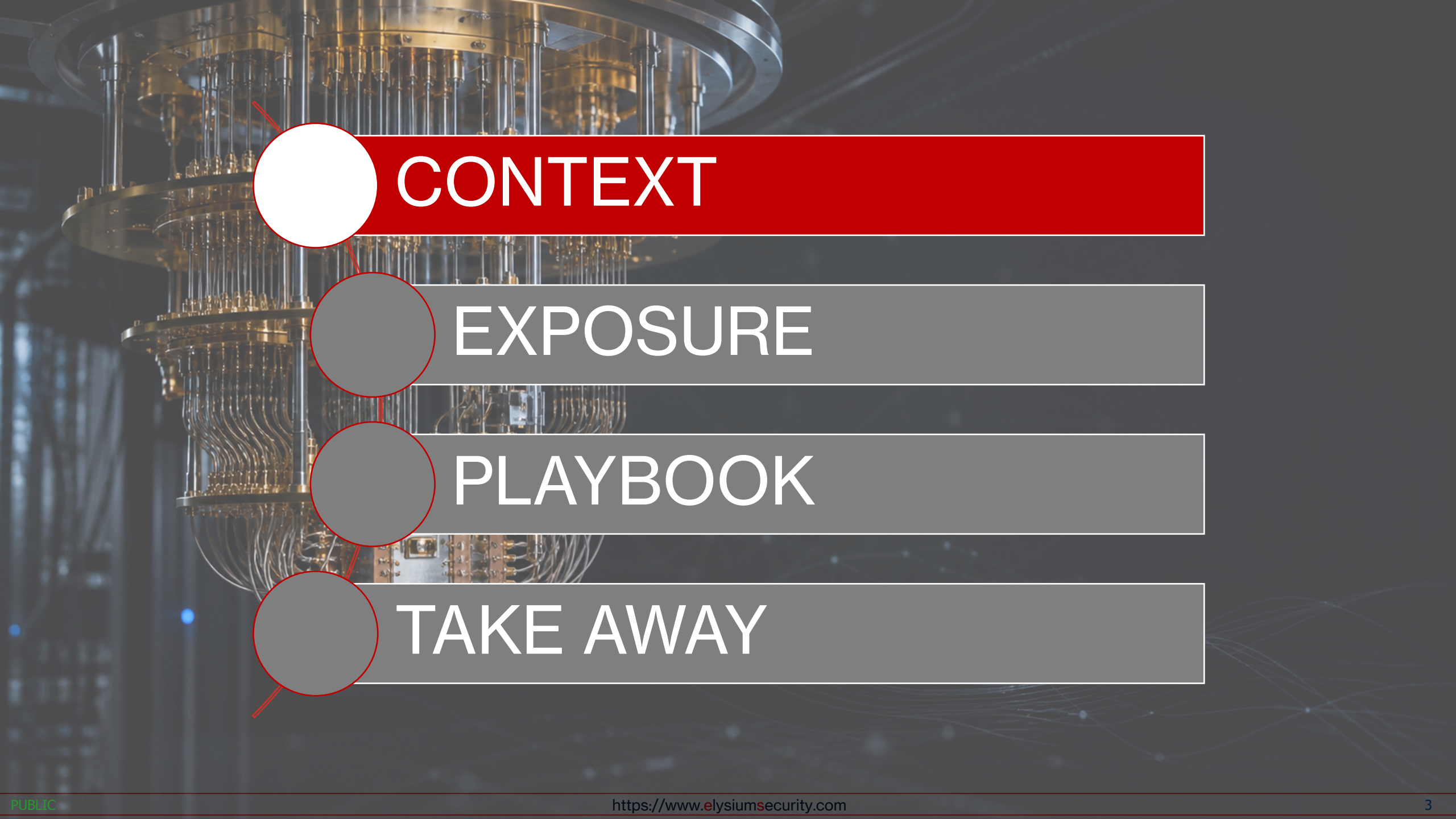
   WHERE VULNERABLE CRYPTOGRAPHY HIDES

3. PLAYBOOK:

   WHAT TO DO, HOW AND WHEN

4. TAKE AWAY:

   WHAT GOOD LOOKS LIKE


CONTEXT
EXPOSURE
PLAYBOOK
TAKE AWAY

PROBLEM STATEMENT: Q-DAY

"QUANTUM COMPUTERS WILL BREAK ALL ENCRYPTIONS,
THERE WILL BE NO SECRETS ANYMORE,
COMMUNICATIONS AND PASSWORDS WILL BE EXPOSED"

THIS WILL BE QUANTUM DOOM'S DAY: **Q-DAY**

Before you panic, remember Year-2000, the supposedly "end of computer world"
(a lot of work was still required though, that's the point!)

FOUNDATIONS – CRYPTOGRAPHY

Cyber Cryptography creates trust in untrusted places



CONFIDENTIALITY

Only Authorized parties can read the information



INTEGRITY

Undetected changes to data become difficult and noisy



AUTHENTICATION

A person, device or service can prove its identity or origin



NON-REPUDIATION

Digital signatures can support evidence that an action occurred

FOUNDATIONS – CRYPTOGRAPHY

There are three cryptographic tools families, each used for different purposes

SYMMETRIC

(PRIVATE KEY ALGO)

One shared Secret key

Fast protection for bulk data
Ex: AES, DES, 3DES, BUGS, etc

ASYMMETRIC

(PUBLIC KEY ALGO)

Public key + Private key

Establish a secret or sign
Ex: RSA, ECC, etc.

HASH

(HASHING ALGO)

No decryption key, one way

A fixed-length fingerprint
Ex: SHA-3, SHA-2, MD5, etc.

For example, a secure web session:

1. Authenticates and agrees keys (ASYMMETRIC)
2. Encrypts the traffic (SYMMETRIC)
3. Hashes data for integrity (HASH)

FOUNDATIONS – QUANTUM

CLASSICAL BIT

0 or 1

A stable state that can be copied and read directly.

QUANTUM BIT - QUBIT

$$\alpha|0\rangle + \beta|1\rangle$$

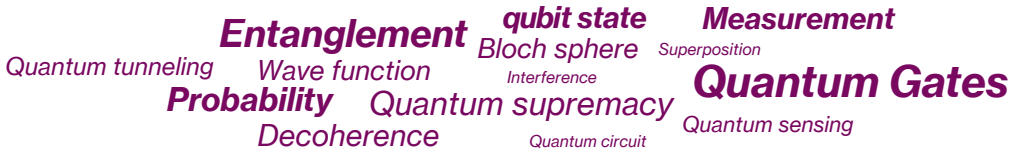
SUPERPOSITION

Gates change probability amplitudes.

INTERFERENCE

The algorithm amplifies useful outcomes.

A QUBIT carries amplitudes and measurement returns a bit (still a classical 0 or 1)



BITS power everyday computing; QUBITS can accelerate specific complex problems.

Because QUBITS can work with multiple possibilities at once

Quantum Computing Explained - https://www.nist.gov/quantum-information-science/quantum-computing-explained

FOUNDATIONS – QUANTUM ALGORITHMS

Quantum computing is not a universal skeleton key.
The effect depends on the algorithm being attacked.

SHOR'S ALGORITHM

PUBLIC-KEY

Efficiently solves the factoring and discrete-logarithm problems used by RSA, Diffie–Hellman and elliptic-curve cryptography.

IMPACT: replace mathematical assumptions

GROVER'S ALGORITHM

PRIVATE-KEY & HASH

Provides a quadratic speed-up for brute-force search. Symmetric encryption and hashes lose security margin, but are not affected like RSA or ECC.

IMPACT: review key and hash sizes

FOUNDATIONS – QUANTUM SPECIFIC TERMS

QUANTUM COMPUTING

Computers that exploit superposition, interference and entanglement.
They create the future cryptographic threat.

CRQC / Q-DAY

Cryptographically Relevant Quantum Computer (CRQC): a sufficiently capable machine.
Q-Day: the point when that capability can attack deployed public-key cryptography.

POST-QUANTUM CRYPTOGRAPHY (PQC)

New mathematical algorithms that run on ordinary computers and aim to resist classical and quantum attacks.

QUANTUM CRYPTOGRAPHY / QKD

Quantum signals, often photons, used to distribute keys and detect interception.
Requires a separate network architecture.

This talk is about migrating ordinary IT systems to PQC and not installing a quantum communications network!

FOUNDATIONS – OVERALL KEY TERMS

KEY

A secret value that controls a cryptographic operation.

KEM

A key-encapsulation mechanism establishes a shared secret over an untrusted network.

DIGITAL SIGNATURE

Uses a private key to support origin and integrity checks, not confidentiality.

CERTIFICATE / PKI

Binds a public key to an identity through a chain of trust.

HYBRID

Combines classical and post-quantum components during transition.

CRYPTO AGILITY

The capability to replace cryptography without breaking security or operations.

SOFTWARE PROGRESS FASTER THAN HARDWARE

NUMBER OF QUBITS REQUIRED TO BREAK RSA-2048 BIT INTEGERS

2019

20 Million
Noisy QUBITS

Time required about 8 hours

Gidney–Ekerå estimate paper
https://arxiv.org/abs/1905.09749

20x fewer QUBITS required in 6 years

2025

<1 Million
Noisy QUBITS

Time required about 1 week

Gidney estimate paper
https://arxiv.org/abs/2505.15917

Not one of those reductions came from a bigger chip.
They came from better mathematics.

QUANTUM CRYPTO STATUS TODAY

156
PROGRAMMABLE QUBITS

Noisy & error prone
IBM Geron processes available through a public cloud

70
LOGICAL QUBITS

Reliable (error correction)
temporary circuit sampling test by IBM in July 2026

21
LARGEST FACTORED NUMBER

Using Shor Algorithm
on a quantum computer
15 in 2001 (IBM)
21 in 2012 (Photonics Paper)

0

Publicly demonstrated Cryptographically Relevant Quantum Computers

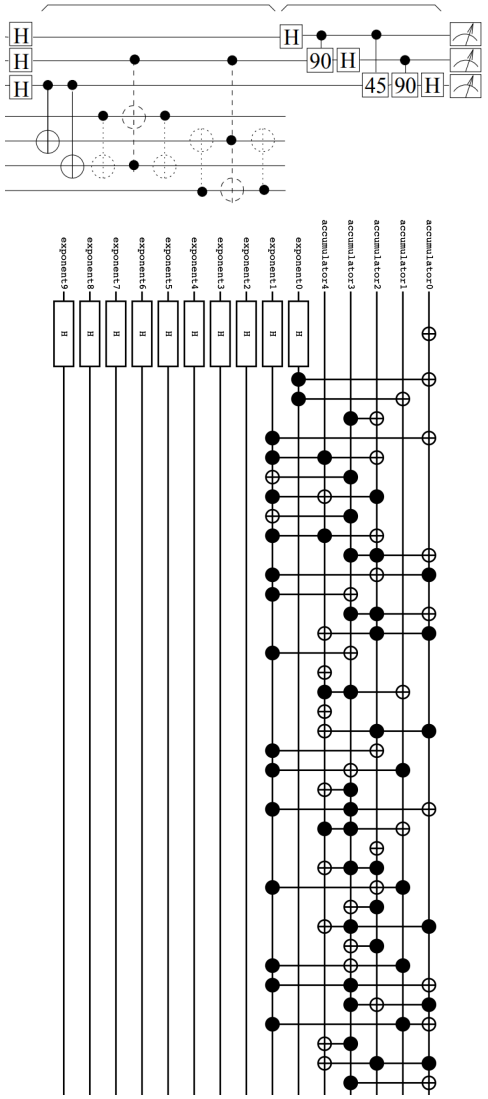
3

Finalized NIST PQC standards ready for implementation

NOW

The recommended time to inventory and plan

No public CRQC exists.
The migration case still stands



OPENQASM2 version of the circuit
Source: https://algassert.com/post/2500

THE PROBLEM: HARVEST NOW, DECRYPT LATER

01

CAPTURE

Encrypted traffic, backups or archives are collected today.

Network taps, transit and subsea cable providers, stolen backups, exfiltrated mailboxes, cloud snapshots, etc.

02

STORE

No decryption is needed yet; storage preserves the option.

Storage is the cheapest thing in computing, and getting cheaper. No alert raised. No IOC exists for a file sitting on someone else's disk.

03

DECRYPT

A future capability is applied only to data that still has value.

Migrating your live systems afterwards does not protect the copies already taken.

Your SSH client already tells you this is happening

```
"** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
```

OpenSSH 10.1 – released October 2025, warning on by default.

WHEN IS IT TOO LATE? ONE FORMULA!

X **SECRECY SHELF LIFE**
How many years this data must stay confidential
No control (law, contracts, etc) but can measure

Y **MIGRATION TIME**
How many years to actually replace the cryptography
Full control and ownership

Z **TIME TO A CAPABLE MACHINE**
Years until a (CRQC) Cryptographically Relevant Quantum Computer exists
No control

"Mosca's Inequality"



EXAMPLE: A HOSPITAL

X **Patient records:** **10-20** years of required confidentiality depending on the country

Y **Realistic migration** of a hospital estate: **5-10 years**

Z **Expert view**: CRQC will exist within **10 years**

10 + 5 = 15

15 > 10

NIST STANDARDS AVAILABLE

Use existing standards, supported protocols and validated products

ML-KEM

(FIPS 203)

KEY ESTABLISHMENT

Establishes shared secrets; it does not sign.
RSA replacement.

ML-DSA

(FIPS 204)

DIGITAL SIGNATURES

General-purpose post-quantum signatures.
Fast and versatile.

SLH-DSA

(FIPS 205)

DIGITAL SIGNATURES

Hash-based alternative with larger signatures.
Heavy and Fall back

→ Linked to NIST SP 800-227 (general guide for KEM)

LATEST NEWS: AI BROKE A PQC CANDIDATE

28 JULY 2026

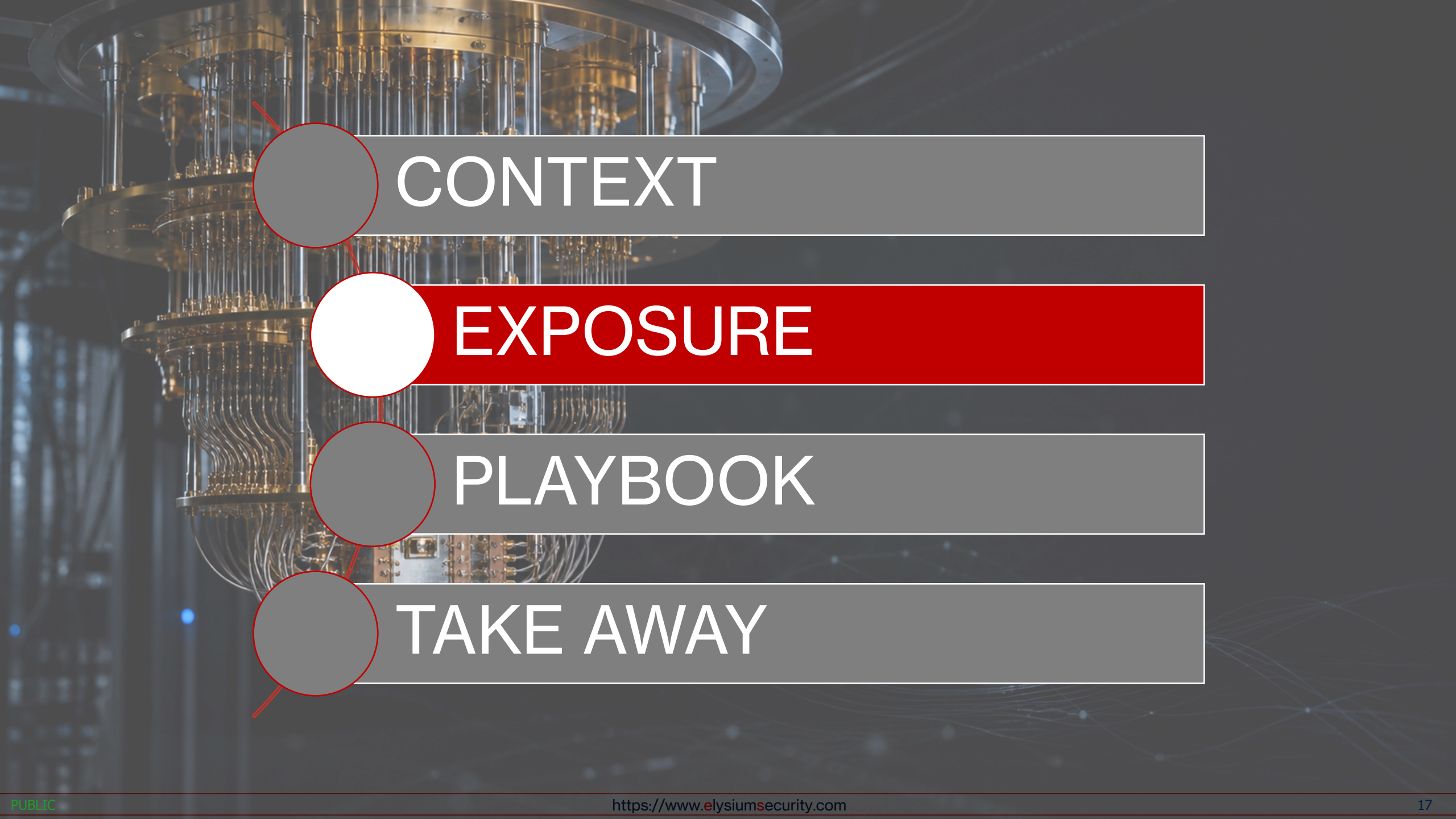
HAWK

A signature candidate was withdrawn after an AI-assisted mathematical vulnerability was disclosed.

WHAT THIS MEANS

- The review process worked: a candidate was removed before standardization.
- NIST says the finalized standards were not affected.
- Your design must support change without a new multi-year project.

THE DURABLE CONTROL IS CRYPTO AGILITY.


CONTEXT
EXPOSURE
PLAYBOOK
TAKE AWAY

EXPOSURE WHAT WILL BREAK AND WHAT WILL NOT

BROKEN BY SHOR'S ALGORYTHM

REPLACE

- RSA - every key size, encryption and signing
- Finite-field Diffie–Hellman / DSA
- ECC: ECDH, ECDSA, EdDSA

ONLY WEAKEN BY GROVER

KEEP & RESIZE

- Symmetric Encryption - AES-256
- Hash Functions - SHA-384/512
- Use policy-approved parameters

SO WHAT DOES THAT MEAN IN YOUR ESTATE?

TLS / HTTPS	Certificates & internal PKI	Secure email (S/MIME, PGP)
VPN & IPsec tunnels	Code and firmware signing	Database & backup key wrapping
SSH and file transfer	Smart cards, Bank payment cards, passports	Blockchain wallets and ledgers
Wi-Fi enterprise auth	HSMs and key ceremonies	IoT, OT and metering devices

'Encryption' will not break soon. Key exchange and signatures might do.
Your AES-256 backup is fine. The key that unlocked it and travelled over RSA, is not!

EXPOSURE YOU ARE ALREADY (PARTIALLY) POST-QUANTUM

ALREADY DONE FOR YOU, FREE

- **Chrome, Edge and Firefox:** hybrid X25519 + ML-KEM by default
- **Apple iOS 26** (Sept 2025): quantum-secure TLS on by default
- **OpenSSH 10.0** (Apr 2025): post-quantum key exchange default
- **Signal:** post-quantum ratchet (SPQR) shipped October 2025
- **Cloudflare, AWS, Google and Meta edges**: hybrid TLS enabled
- **Result:** 1/3 of human web traffic in Jan 2025 -> 2/3 today

NOT DONE FOR YOU

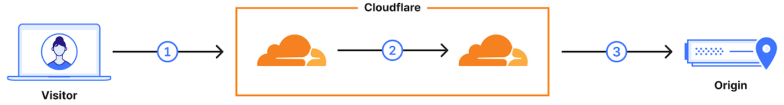
- Site-to-site VPNs, firewalls and SD-WAN
- HSMs, key ceremonies, internal CA and certificate estate
- Code signing, firmware signing, software supply chain
- SFTP / MFT feeds to banks, regulators and partners
- Database, backup and archive key wrapping
- ATMs, POS, smart cards, badges, smart meters, etc.

TEST IT YOURSELF

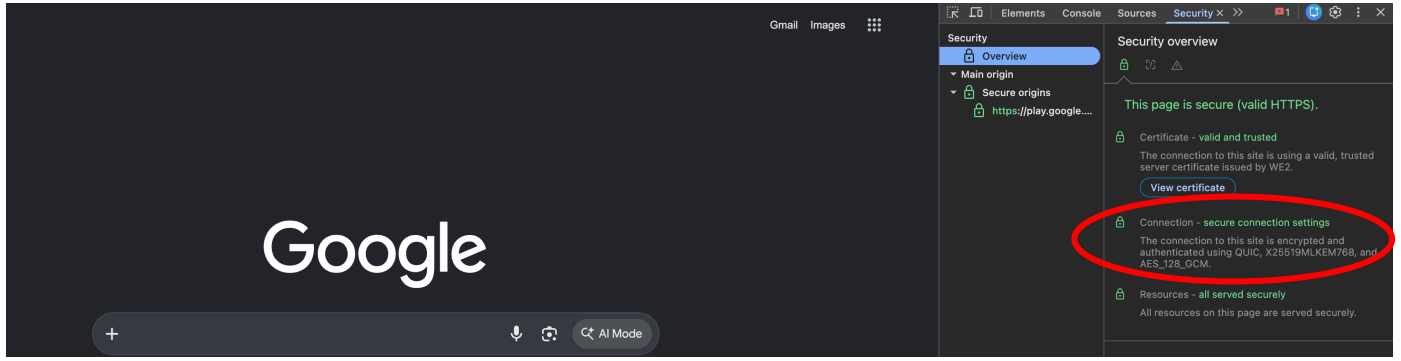
Your Browser: https://pq.cloudflareresearch.com/

The website you visit

Cloudflare Research: Post-Quantum Key Agreement



You are using *X25519MLKEM768* which is **post-quantum secure**.



EXPOSURE DIFFERENT CLOCKS

CONFIDENTIALITY

DEADLINE: TODAY

- Attackable retroactively through collected traffic or archives.
- Prioritise long-lived data and exposed transport paths.
- Hybrid key exchange may reduce risk where supported and justified.

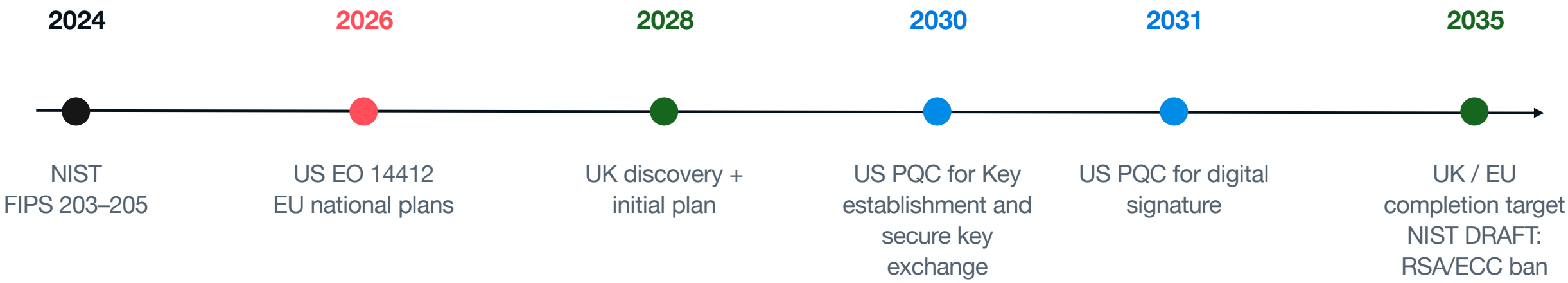
AUTHENTICITY

DEADLINE: TRUST CYCLE

- A future forgery matters while the verifier still trusts the classical key.
- Roots, HSMs, firmware, devices and smart cards create long lead times.
- Plan signature migration around renewal and technology-refresh cycles.

Confidentiality is a race against a machine.
Authenticity is a race against your own renewal cycle.

MIGRATION TIMELINE AND DEADLINE



Jurisdiction-specific policy signals – not a universal compliance timeline.

Common message: Inventory -> Priorities -> (Supplier) plans -> Multi-year execution

EXPOSURE MAURITIUS AND THE INDIAN OCEAN

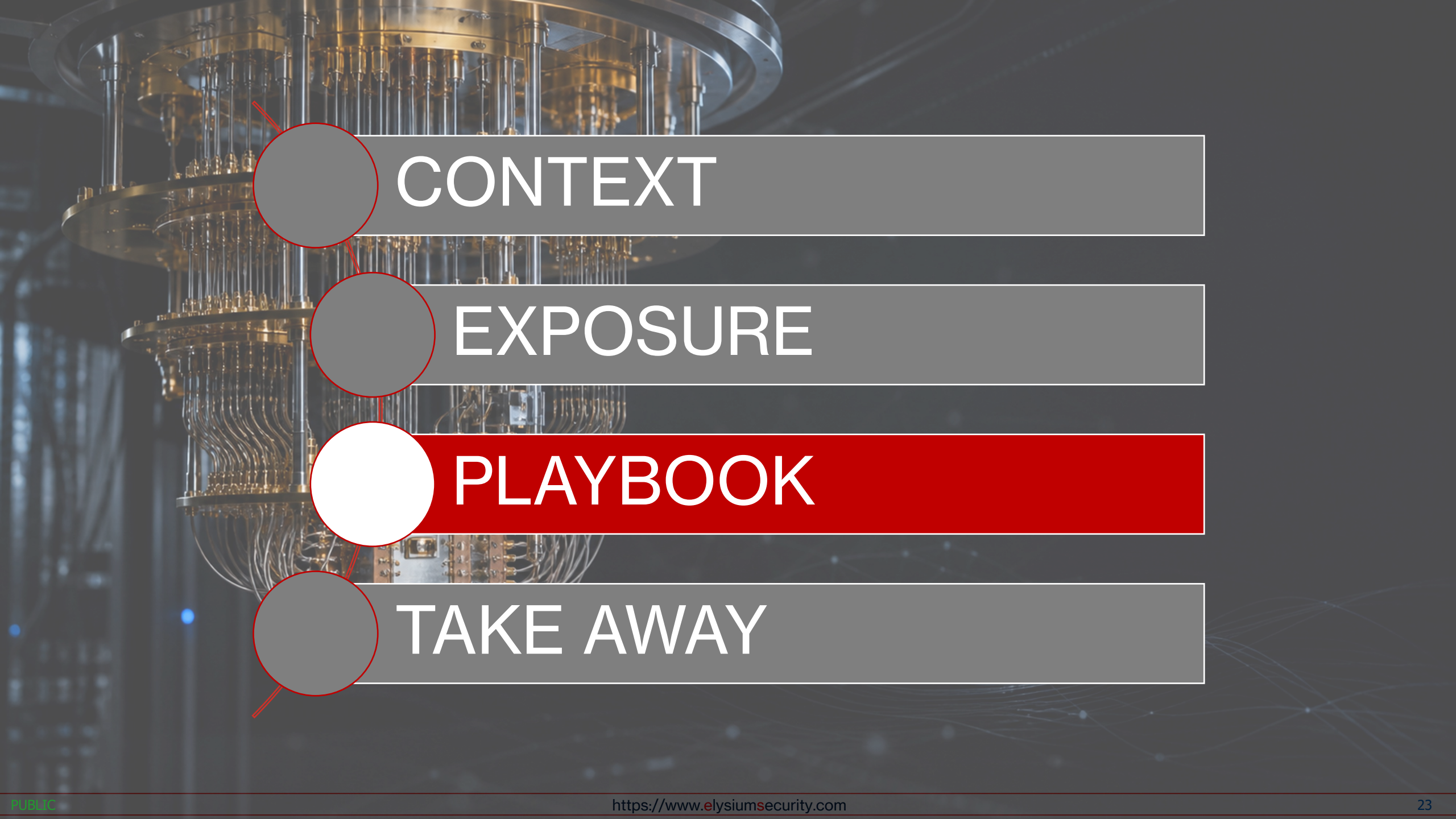
THE RULES THAT ALREADY MOVED

- Bank of Mauritius Guideline on Cyber and Technology Risk Management (effective 29 May 2023), s.28(i): cryptographic controls must follow "*recognised standards and processes... algorithm, key length, and key generation*"
- Since 13 August 2024 the recognised standard includes ML-KEM and ML-DSA.
- Data Protection Act 2017: "*appropriate technical measures*" are judged against the risk. For harvested data, that judgement lands in the year it is finally decrypted.

HOW THE DEADLINE WILL ARRIVE HERE

- Parents company in Europe, India and South Africa exporting policy
- International banks passing their own 2030 obligations down the chain
- EU clients under NIS2 and DORA assessing you as a third-party supplier
- Card schemes and payment processors updating technical requirements
- Cloud and SaaS providers changing defaults and deprecating old suites

The deadline may not first arrive from a regulator.
It will likely arrive in a supplier questionnaire.


CONTEXT
EXPOSURE
PLAYBOOK
TAKE AWAY

PLAYBOOK START WITH 5 QUESTIONS TO ANSWER

1. **What cryptography do you actually run?**

   Algorithms, key lengths, where the keys live, and which vendor owns each one. Not a diagram, a list.

   → *Most important point*

2. **What is your longest-lived secret, and when does it stop mattering?**

   One line per data type. The answer decides your priority order, not the age of the system.

3. **Who owns this?**

   Name a person, not a department. A migration without an owner is a slide deck.

   → *usually owns by nobody today*

4. **What have your vendors committed to, in writing, with a date?**

   Verbal reassurance from an account manager is not a roadmap. Ask for the release and the version.

5. **Could you change an algorithm without running a project?**

   If changing a cipher requires a business case, you do not have a migration problem: you have an agility problem.

NO/LOW COST

→ YEARS TO ACHIEVE

PLAYBOOK STEP 1: DISCOVERY

NETWORK EDGE

TLS versions and cipher suites, load balancers, reverse proxies, VPN and IPsec, SD-WAN, Wi-Fi enterprise auth.

CERTIFICATES & PKI

Internal CA, root and issuing key algorithms, anything valid beyond 2030, certificate discovery across the estate.

CODE & PIPELINES

Hardcoded algorithms and key sizes, crypto libraries and pinned versions, signing and release pipelines.

KEYS, HSMs & KMS

Key wrapping, escrow, ceremonies, cloud KMS defaults, which HSM firmware supports which algorithms.

VENDORS & SAAS

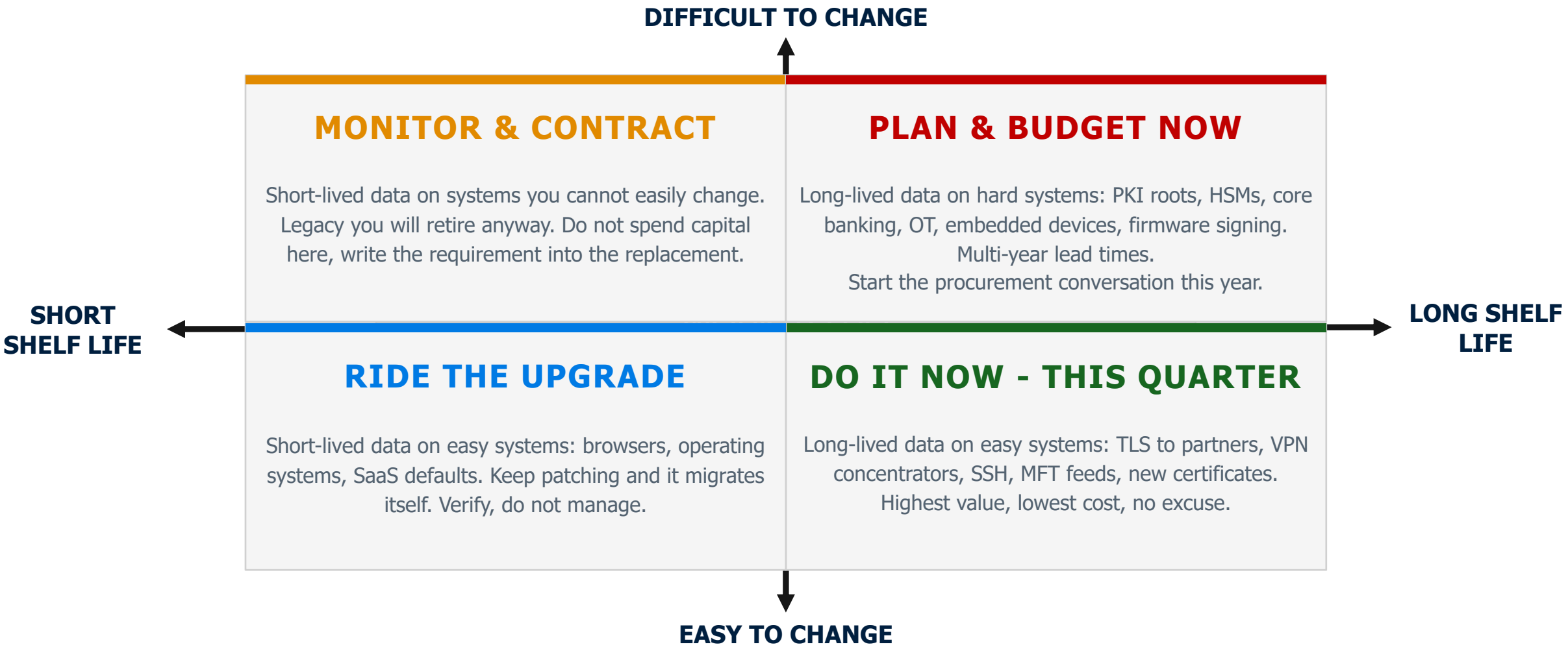
Every supplier's stated algorithms, roadmap dates and version numbers: captured in the contract file, not in an email.

OT, IOT & CARDS

Firmware signing, ATMs, POS terminals, smart cards, badges, meters, medical devices, anything with a 10-year life.

You cannot migrate what you cannot list.
A simple spreadsheet done within a month will save you future expenses

PLAYBOOK STEP 2: PRIORITISATION



Start bottom-right. It is the cheapest quadrant, and it closes the harvesting window first.

STEP 3: TURN ON WHAT IS ALREADY FREE

1	Enable hybrid post-quantum key exchange at your CDN, edge and load balancers, usually a single setting.
2	Upgrade to OpenSSH 10.x and stop dismissing the "store now, decrypt later" warning.
3	Push browser and operating system updates, a large share of your exposure closes by itself.
4	Ask your VPN, firewall and MFT vendors for their post-quantum release. Several already ship it and nobody enabled it.
5	Move symmetric cryptography to AES-256 and SHA-384 wherever it is a configuration option.
6	**Stop issuing long-lived RSA-2048 certificates, and stop signing roots that expire after 2030 without a plan.**
7	Add "post-quantum roadmap, with version numbers and dates" to your standard supplier questionnaire.

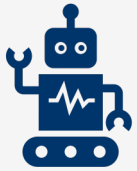
Doing these seven closes the harvesting window on most of your traffic before your budget cycle even opens.

PLAYBOOK STEP 4: CRYPTO AGILITY IS KEY



ABSTRACT

No hardcoded algorithm names or key sizes in business logic.



AUTOMATE

Issue, rotate and revoke keys and certificates without tickets.



OBSERVE

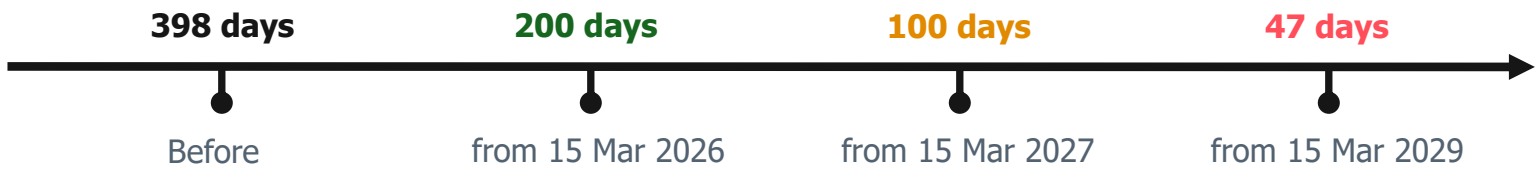
Know what is negotiated, where it failed and what remains exposed.



CHANGE AGAIN

Maintain inventory, rollback and supplier evidence as standards evolve.

PUBLIC TLS CERTIFICATE MAXIMUM VALIDITY EXAMPLE



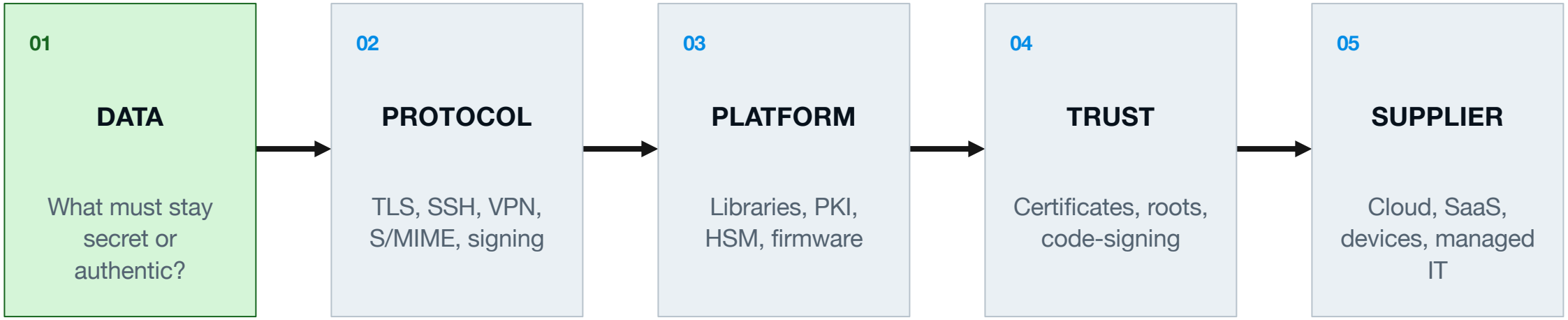
There will be more than one migration. Build the ability to change, not a single change.

STEP 5: CONTRACT CHALLENGE

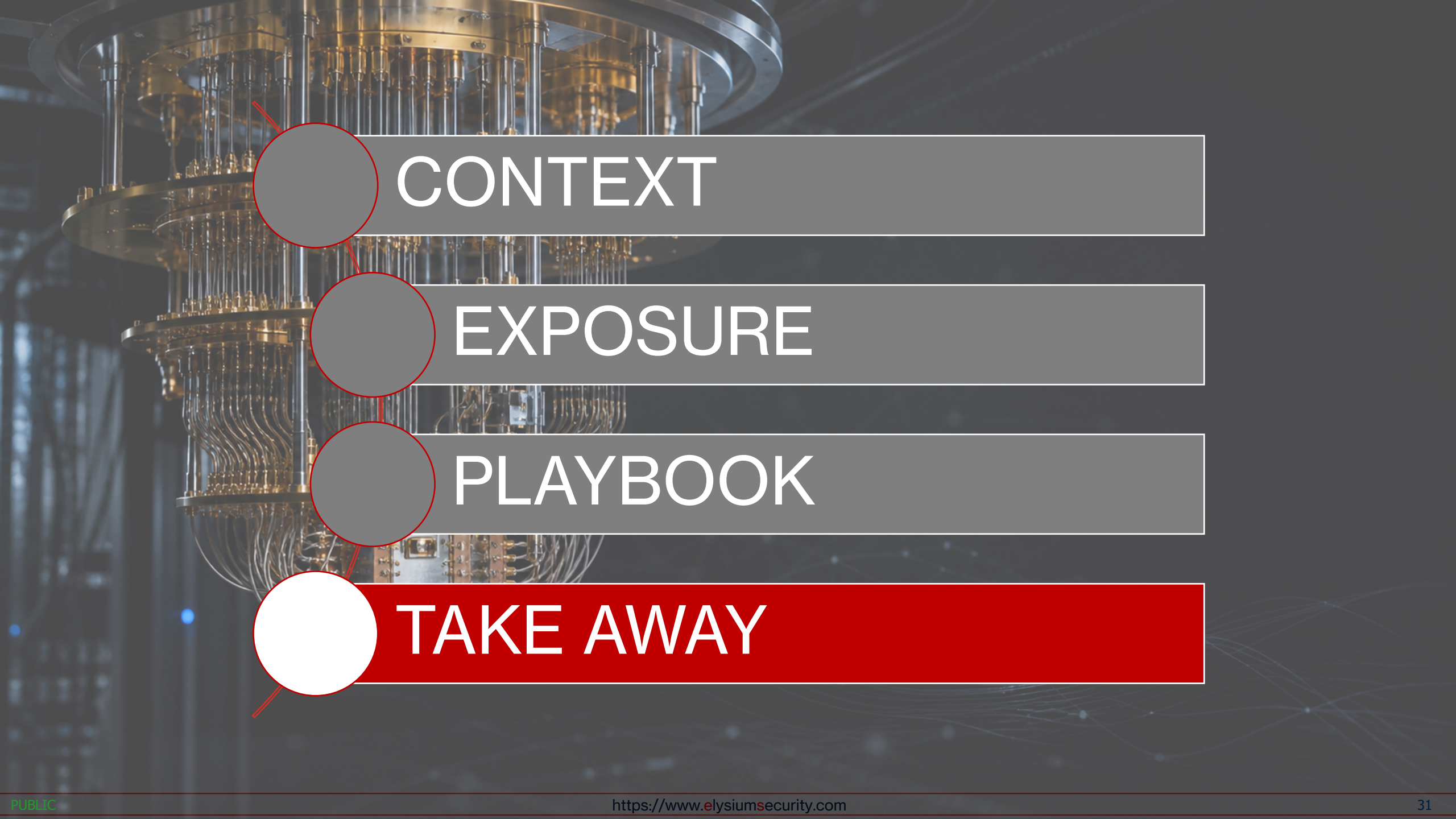
SUPPORTED NOW	Which NIST algorithms, parameters and protocols are in which product version today?
ROADMAP	Separate dates for hybrid key exchange, post-quantum signatures and validation.
CHANGEABILITY	Can the algorithm change without replacing hardware or breaking the service contract?
DATA ALREADY HELD	How are our backups, archives, keys and replicas protected through the transition?
EVIDENCE	What testing, interoperability, rollback and certification evidence will you provide?

Do not buy 'quantum-safe'
Buy a supported version, a date, a test plan and a rollback path.

MIGRATION PROCESS: DEPENDENCY CHAIN



Change any one link without testing the others and the service may still be vulnerable, or stop working


CONTEXT
EXPOSURE
PLAYBOOK
TAKE AWAY

TAKE AWAY WHAT GOOD LOOKS LIKE

OWN	KNOW	CHANGE	PROVE
A named risk owner and decision forum.	A prioritised and maintained inventory linked to business services	Tested migration, automation and rollback paths.	Supplier commitments, metrics and evidence.

Four repeatable capabilities
Three are **governance** capabilities. Only one is primarily **cryptographic**.

THE OBJECTIVE IS NOT TO BE QUANTUM-SAFE BY FRIDAY
Know what you run. Protect what must still be secret in 2035. Be able to change again.

TAKE AWAY 5 ACTIONS IN THE NEXT 4 MONTHS

THIS MONTH	**Name the owner.** One person, accountable, on the risk register. Not a department, not a committee.
NEXT MONTH	**Start the inventory.** External TLS scan, certificate discovery, and a questionnaire to your ten most important suppliers.
THIS QUARTER	**Turn on the free wins.** Edge and load balancer hybrid TLS, OpenSSH 10.x, OS and browser updates, AES-256 where it is a setting.
ONE MEETING	**Write down your three longest-lived secrets.** And the year each one stops mattering. That single page sets every priority that follows.
FROM NOW ON	**Put a product version and dates in every contract.** Post-quantum roadmap, version number and date: in every renewal, RFP and third-party assessment.

Nobody can finish this migration in 2026.
Everybody can start it in 2026.

SOURCES AND FURTHER READING

STANDARDS AND TECHNICAL

- NIST FIPS 203 / 204 / 205 (13 Aug 2024) — ML-KEM, ML-DSA, SLH-DSA
- NIST IR 8547 (draft) — transition plan: deprecate 2030, disallow 2035
- NIST IR 8545 — HQC selection, March 2025
- NIST SP 800-227 — key-encapsulation mechanisms, Sept 2025
- Gidney, arXiv:2505.15917 — RSA-2048 with under a million qubits
- Cain et al., arXiv:2603.28627 — Shor with ~10,000 atomic qubits

POLICY, REGULATION AND PRACTICE

- US Executive Order 14412 (22 June 2026)
- EU Coordinated Implementation Roadmap for PQC (June 2025)
- UK NCSC — timelines for migration to post-quantum cryptography
- NSA CNSA 2.0 — national security systems timeline to 2033
- Global Risk Institute — Quantum Threat Timeline Report 2025
- Federal Reserve FEDS 2025-093 — harvest now, decrypt later
- Cloud Security Alliance — AI infrastructure and HNDL (May 2026)
- Bank of Mauritius — Guideline on Cyber and Technology Risk Management

THANK YOU! Any Questions?



ABOUT ELYSIUMSECURITY LTD.

ELYSIUMSECURITY PROVIDES A PORTFOLIO OF **STRATEGIC** AND **TACTICAL SERVICES** TO HELP COMPANIES **PROTECT** AND **RESPOND** AGAINST **CYBER SECURITY THREATS**. WE DIFFERENTIATE OURSELVES BY OFFERING **DISCREET**, **TAILORED** AND **SPECIALIZED** ENGAGEMENTS.

ELYSIUMSECURITY OPERATES IN **MAURITIUS** AND IN **EUROPE**, A ***BOUTIQUE STYLE*** APPROACH MEANS WE CAN **EASILY ADAPT** TO YOUR BUSINESS OPERATIONAL MODEL AND REQUIREMENTS TO PROVIDE A **PERSONALIZED SERVICE** THAT FITS YOUR WORKING ENVIRONMENT.

ELYSIUMSECURITY PROVIDES **HIGH LEVEL EXPERTISE** GATHERED THROUGH YEARS OF **BEST PRACTICES EXPERIENCE** IN LARGE INTERNATIONAL COMPANIES ALLOWING US TO PROVIDE ADVICE BEST SUITED TO YOUR **BUSINESS OPERATIONAL MODEL** AND **PRIORITIES**.

ELYSIUMSECURITY PROVIDES **PRACTICAL EXPERTISE** TO **IDENTIFY VULNERABILITIES**, ASSESS THEIR **RISKS** AND **IMPACT**, **REMEDIATE** THOSE RISKS, **PREPARE** AND **RESPOND** TO INCIDENTS AS WELL AS RAISE **SECURITY AWARENESS** THROUGH AN ORGANIZATION.

